LAB 1: REQUIREMENTS ANALYSIS & USE-CASE MODEL

Restaurant Table Booking & Pre-Ordering App

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GitHub Repo: https://github.com/SakshamBajpayi/SE-lab-PES1UG24CS404

1. Functional Requirements

These requirements cover the main things the app needs to do, like letting users pick tables, order food from the menu, and handle kitchen timing. It also includes how the manager can control which tables are available for booking.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Booking	The system shall allow diners to pick an open table from an interactive 2D floor plan and link pre-ordered menu items to the reservation.	High	Pass: Table reserved and kitchen pre-order ticket queued. Fail: Overbooking of table during overlapping dining window.	Essential core feature linking floor plan reservation with food pre-ordering.
FR-002	Menu & Order	The system shall display a real-time menu allowing diners to select items, specify quantities, and add dietary preferences for pre-ordering.	High	Pass: Items successfully added to cart with custom notes. Fail: Out-of-stock items can still be added to the order.	Diners need accurate menu information to place advance food orders successfully.
FR-003	Prep Timeline	The system shall calculate the kitchen preparation timeline based on the pre-ordered items and reservation time, scheduling the kitchen ticket accordingly.	High	Pass: Kitchen ticket is scheduled such that prep finishes within 5 minutes of diner's arrival. Fail: Prep time is scheduled without considering standard dish preparation durations.	Ensures seamless arrival dining by having food ready exactly when the diner is seated.
FR-004	Notification	The system shall send a reservation confirmation including the selected table number, pre-order summary, and a QR code for check-in to the diner's email/app.	Medium	Pass: Diner receives confirmation with all details within 2 minutes of booking. Fail: Notification fails to send or omits the table number/pre-order details.	Provides diners with proof of reservation and details needed for quick seating upon arrival.

FR-005	Floor Plan Update	The system shall allow the Restaurant Manager to manually update table availability (e.g., block tables for maintenance) on the interactive 2D floor plan.	Medium	Pass: Blocked table immediately shows as unavailable to diners. Fail: Diners can still reserve a manually blocked table.	Managers need control over the floor plan to handle real-world dining constraints.

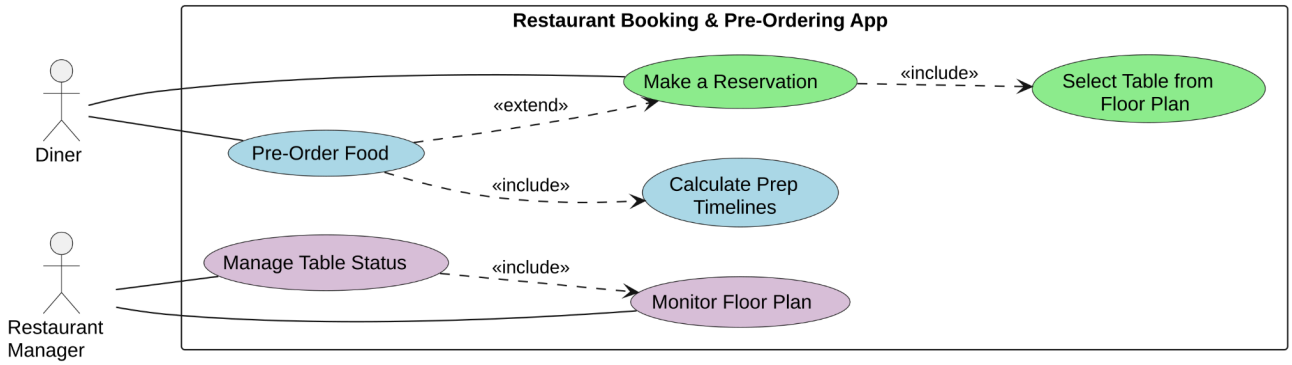
2. Non-Functional Requirements

These are the quality and technical constraints for the app, focusing on how fast things sync up, general performance, security, and making sure the mobile version works well.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Table status changes (Reserved, Seated, Billed) must sync across all manager floor terminals in under 500 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Prevents double-booking and ensures staff have the most up-to-date view of the restaurant floor.
NFR-002	Usability	The interactive 2D floor plan must be fully responsive and render accurately on mobile devices (iOS and Android) with a minimum touch-target size of 44x44 pixels.	High	Pass: User testing verifies floor plan usability and accurate table selection on mobile screens. Fail: Elements overlap or touch targets are too small on mobile.	Diners will predominantly use the hospitality app on mobile devices, requiring a seamless mobile experience.

3. Use-Case Model

This diagram shows how the Diner and the Restaurant Manager use the app. The Diner mainly handles bookings and ordering food, while the Manager takes care of table statuses and keeps an eye on the floor plan.



4. Use-Case Relationships

Relationship	Use Cases	Meaning in the Diagram
«include»	Make a Reservation → Select Table from Floor Plan	Selecting a table from the floor plan is included in the reservation flow.
«extend»	Pre-Order Food → Make a Reservation	The pre-ordering behavior extends the reservation scenario.
«include»	Pre-Order Food → Calculate Prep Timelines	Preparation timeline calculation is included in the pre-ordering flow.
«include»	Manage Table Status → Monitor Floor Plan	Monitoring the floor plan is included in the manager's table-status management flow.

5. Actor Responsibilities

Actor	Associated Use Cases
Diner	Make a Reservation; Pre-Order Food
Restaurant Manager	Manage Table Status; Monitor Floor Plan

6. Requirements-to-Use-Case Traceability

Requirement	Related Use Case(s)	Traceability
FR-001 Booking	Make a Reservation; Select Table from Floor Plan	Directly represented by the Diner reservation flow and its included table-selection behavior.
FR-002 Menu & Order	Pre-Order Food	Represented by the Diner's pre-ordering use case.
FR-003 Prep Timeline	Calculate Prep Timelines	Explicitly represented as an included use case in the diagram.
FR-004 Notification	No explicit notification use case shown	The functional requirement exists in the requirements table, but the uploaded use-case diagram does not show a separate notification use case.
FR-005 Floor Plan Update	Manage Table Status; Monitor Floor Plan	Related to the Restaurant Manager's floor-plan and table-status responsibilities.
NFR-001 Performance & Security	Applies system-wide	Quality constraint rather than a separate user-facing use case.
NFR-002 Usability	Applies primarily to floor-plan interaction	Quality constraint affecting the interactive 2D floor plan and mobile experience.